Orion: the whole application

The internal workspace of Prakash Securities.

What it is, how it is built, and what each part does.



What Orion is

Run the business

Clients, cases, employees, vehicles, tenders and their earnest money. Attendance that turns into pay.

Know the money

Tax invoices, payments, TDS, credit notes, GST, expenses, salary, and a bank balance that is calculated, never typed.

See the picture

Eight dashboard tabs, a forecast, and a read-only copy of the Tally books reported side by side.

Work together

Tasks, chat and calls, a Document Center, letters on the letterhead, and an AI Copilot built into every screen.

FastAPI backend · React frontend · one PostgreSQL database · one Windows PC

Orion by the numbers

58,000

lines of Python in the backend

181 files, 37 API routers

71,000

lines of TypeScript in the frontend

356 files, 79 pages

45

database schema files

applied in order, safe to re-run

1,656

automated tests pass

72 test files, no database needed

118

pull requests merged

never a commit straight to main

265

commits

since 31 August 2026

14

kinds of business record

forms, Excel upload and the Copilot

8

dashboard tabs

plus a forecast and the Tally dashboards

The system in one picture

CLIENT

Browser or desktop window

React app. One WebSocket for chat; a session cookie.

HTTP →

FastAPI (backend/app)

- Routers: thin, one permission check each
- Modules: the logic and the SQL, no ORM
- Schema applied at every start
- A pool of database connections
- Threads: warehouse auto-load, backups, tender checks, document filing, repeating tasks

→

DATABASE

PostgreSQL, one database

orion_ods: the records the app writes.
orion_dw: versioned warehouse.
tally_ods and tally_dw: the Tally books.

→

Tally Prime

XML server on port 9000, office PC only. Read only.

Where it runs

Development: UI on port 5173, API on 8000, database "orion". Production, on the office PC only: the API serves the built UI on port 8100, database "orion_prod", in a folder that only the deploy script changes. A native-window launcher opens either one.

How a change reaches the office PC

Branch

Never a commit to main. Work on a branch and push it.

Pull request

Reviewed and merged on GitHub.

CI

Backend, frontend and dependency checks run on every PR, with no database.

deploy.ps1

Safety dump, then migrations, then stop, update, start.

Smoke test

The new server is tested. If it fails, the old version returns.

Schema changes are additive

Every SQL file is safe to re-run. Nothing is dropped or renamed in the release that stops using it. A failed migration stops the deploy.

Rollback is built in

The previous version is noted first. If the smoke test fails, it is put back. /api/health/version reports the running commit.

Rules the agents follow

AGENTS.md and change-impact.md list what else breaks when something changes, and what a change must prove before it is called done.

Four schemas, each with one job

orion_ods

The books of record

Everything the app creates, changes and deletes: bills, expenses, employees, users, chat. The only schema the app writes.

orion_dw

Orion warehouse

A versioned copy (SCD Type 2) of the records. Where history lives. The Orion dashboards read it.

tally_ods

Tally working copy

The accountant's books, copied from Tally Prime. Latest value wins (SCD Type 1).

tally_dw

Tally warehouse

Built only from tally_ods, versioned. The Tally dashboards read it.



loaded hourly if changed



loaded from Tally, then warehoused

The Orion and Tally sides are independent: no table in one references the other.

The screen side: one React app, two looks

One app, lazy pages

- React, TypeScript, Vite, Tailwind 4, Recharts
- Every page loads on demand
- Start-up download kept under 650 kB

Themes as data

- Six themes, light, dark or device
- Accent, style, radius, density, size
- A new theme is one object, checked for contrast

Classic and Lumen

- Same pages, permissions and data
- Classic: side rail and tables
- Lumen: floating dock, glass cards, headlines

One door to the API

- Every call goes through api/client.ts
- Types in types.ts
- The server, not the screen, enforces access

Tables with filters

- Every column has a filter
- Server paging, one safe filter builder
- Money typed with Indian commas, read back in words

Workspaces stay alive

- Five workspaces stay mounted once visited
- Going back keeps scroll, filters, forms
- Heavy chart code stays out of the shell

A map of the application

INSIGHTS

- Home
- Orion dashboards
- Tally dashboards

DATA

- Orion data (records, clients, cases, employees)
- Tally data
- ETL (pipelines)
- SQL editor (admins)

UTILITIES

- Billing (and Outstanding, Credit notes, TDS, GST, Monthly billing)
- Letterhead
- Documents
- Bank
- GeM tenders
- MP tenders
- Attendance

TEAM

- Messages
- Tasks

Everywhere: the Copilot (Ctrl+J), the command palette (Ctrl+K), notifications, a refresh button and a theme menu in the top bar.

Who may do what: a role and access flags

Operational data
ods_access / ods_write

Billing
billing_access / billing_write

Bank
bank_access / bank_write (admins only until given)

ETL pipelines
etl_access / etl_write

Letterhead
letterhead_access / _write

Documents
documents_access / _write

Attendance
attendance_access / _write

Tenders
tenders_access / _write

Dashboards
orion_dashboard_access, tally_dashboard_access

AI Copilot
ai_access (on by default; it does only what the person's other flags allow)

Warehouse, Tally
dw_access, tally_access, tally_dw_access

Admin role
users, reversing a payment, letterhead, purging documents

Safeguards built into the platform



Sessions the server can end

A session is a row in the database. It ends on sign-out, a password change, a disabled account, or an admin's "Sign out everywhere".



Sign-in is rate-limited

Failed attempts are limited per name and address. Every route reachable without signing in is throttled.



Strict page policy

In production the page obeys a Content-Security-Policy: no inline scripts, and only known hosts for scripts, fonts and calls.



Private columns

Aadhaar, PAN and monthly salary are hidden from people who can only view; Aadhaar and PAN never reach the warehouse.



SQL editor is guarded

Admins only, read-only in production by default, audited on every run, and it never reads private tables or changes tables the code owns.



Production refuses dev defaults

A development session secret, database or admin password stops the production start. Backups are encrypted.

Operational Data: fourteen kinds of record

WHO WE SERVE

- Client
- Contact (the people at a client)
- Case (a contract at a site)
- Opportunity (a tender or lead)
- EMD (earnest money deposit)

WHO WORKS FOR US

- Employee
- Employee Transaction (a salary line)
- Car
- Car Transaction (a trip)

MONEY LINES

- Transaction (a revenue line)
- Expense Transaction
- Expense Type

HOW WE GROUP

- Business Domain
- Capability (a service line, with a head)

A case belongs to a client and a capability; costs and revenue lines hang off a case; an employee is worked at a case. Codes such as CL-0007 are generated by the database, never typed.

How the data forms work

One registry, three uses

The same description drives the form, the Excel template and upload, and the Copilot. All writes go through entity_service.py.

Two people, one record

An edit carries the version it was opened at. If someone saved since, it is refused with "changed by X after you opened it".

Who, when, what

created_by and last_updated_by on every write, a log of deletes, and a History view with each field's old and new value.

Totals worked out for you

Amount after tax and GST, driven kilometres and the salary month are computed the same way on a form and in Excel.

Values are checked once

Numbers must be real, dropdowns must match, no NaN. A bad value is a clear 400, never a crash.

Locked records

A revenue line that belongs to a bill, and the lines an attendance register entered, cannot be edited here.

A page for each client, case, employee and account



Client

- Who they are and their group
- The cases we run for them
- The people we deal with
- Billed, received and still owed, with ageing (billing access)



Case

- The site and the people placed there
- Contacts, and the client's main contact
- Deposits (EMDs) and expenses
- Billed and owed (billing access)



Employee

- Case and client worked at
- Who they report to, and who reports to them
- Contacts they look after
- Latest salary lines (bank access)



Account (admins)

- What the account may do, in words
- Can change / Can see / No access
- When and from where it was last used
- The tasks on it

From a tender to a case: the won-opportunity follow-up



Tender found

GeM or MP Tenders search, or entered by hand.



Opportunity

Proposal, Qualified, Technical Bid, Financial Bid.



Closed Won

Saved as won.



The bell asks

Create the case. Record the EMD.



It settles itself

A case or an EMD now exists.

It keeps coming back

The notification stays until the case or EMD exists or a person says "not required" with an optional reason.

Clicking it opens the form

The Add form opens with the opportunity already chosen and fixed. An EMD needs a case, so the case comes first.

EMD tracks the whole life

Amount, instrument, deposit, refund due and received, kept back. The Pipeline tab shows EMD still held, overdue and forfeited.

Billing, bank and attendance

The daily work. The money rules are in the other deck.

Billing: raise, print, send, correct, collect



Raise a bill

- Case, lines, charges, GST
- The server recomputes every figure
- Number FY/seq from a locked counter
- Copy as new, or from Monthly billing



Print and send

- PDF drawn on the server
- On the company letterhead if chosen
- Email with the PDF, or WhatsApp
- Saved to Google Drive by choice



Edit with history

- A reason; a new version
- The old version stays viewable
- Its own typeface and size
- Fixed colours, never the app theme



Collect

- Payments in full, in part, net of TDS
- GST paid in instalments
- Credit notes and write-offs
- Reminders by email or WhatsApp



Find a bill

- Search by number, case, client, site
- Type a month: "sep", "2026-09"
- Filter by client, case and month
- Any year, or one



Safe by design

- A number is never changed or reused
- Saved once, even on a double click
- A bill cannot go below what was received
- Every change leaves a version

The six Billing screens

BILLING

The register: every bill, with filters, and the bill editor and viewer.

OUTSTANDING

What each client owes, aged 0-30, 31-60, 61-90 and over 90 days. Reminders and statements.

CREDIT NOTES

Make, approve and reverse credit notes and write-offs.

TDS

TDS held back, by financial year, client and quarter, to match Form 26AS.

GST

GST on bills, input tax credit and net to pay, month by month.

MONTHLY BILLING

Cases billed before but not yet for the chosen month, each with "Copy last bill".

The Bank: six screens over one calculated balance

OVERVIEW

Balance, this month and year, interest and other income, charts and the cash forecast.

LEDGER

Every movement with the balance after it. Filters, CSV export, add and void entries.

PAYABLES

What is owed on expenses, with ageing, and the payments that settle it.

PAYROLL

Mark a month's salary paid in one pay run: the lines, the day, the account.

RECONCILE

Upload a statement, match it to the book, work the lines that do not match.

ACCOUNT

The opening balance, the minimum balance warning, and the history of what was done.

Several accounts are supported, bank and cash. A statement belongs to one bank account; a cash account is counted, not reconciled.

Attendance: who worked where, and the pay that follows

The register

- One case, one month, a line per person
- Days present, rate for a day, extra, deduction, amount
- Employees counted or left out with a tick
- Excel template, upload and export

Employees and workers

- Employees come from the employee records
- Casual labour is added by name
- Workers are never made employees
- Saved for the next register

Posting

- Confirmation, count and total the person saw
- Employees become salary lines
- Workers become expenses of a chosen type
- Nothing is paid by posting

Reopen

- Until anything it entered has been paid
- Takes the entries back, with a reason
- Every step is kept in its own history
- Post again after correcting

Tenders: GeM and MP Tenders, checked for you



Saved searches

- Tender types (words in the category)
- Up to 100 cities, "All cities of a state"
- Services, products or all; value range
- GeM and MP Tenders are separate tools



Checked by itself

- Every six hours, or "Check now"
- Reads the public bid document
- Value, EMD, closing date, contract period
- Turnover and experience asked for, and place



A person decides

- Shortlist or dismiss each bid
- Filter by closing date, value, EMD, place
- In Lumen, a days-left tile on each bid
- A won bid continues as an opportunity



Polite by rule

- Only pages that ask for no captcha
- Never signs in, never answers a captcha
- A gap between requests
- Stops on a refusal, never retries

Dashboards, forecast and Tally

Turning the records into decisions.

The Orion dashboards: eight tabs

Overview
Revenue, cost, margin, cash and what is owed, with what stands out.

Revenue
Billing by service line, client and state; concentration; contracts billing below expectation.

Collections
Cash against billing, ageing of what is owed, who owes and who is slow.

Profitability
Margin by case, domain, capability, client type and year; the case portfolio; cost mix.

People
Headcount, joiners and leavers, attrition, payroll, span of control.

Pipeline
Tenders by stage, win rate, why bids are lost, what wins now bill, EMD still to come back.

Fleet
Distance, cost, cars, drivers and km allowance.

Controls
Cancelled or edited bills, unpaid GST and salary, unbilled cases, follow-ups due.

A ninth tab, Cash, reads the Bank live and is for people with bank access. A tenth, Forecast, looks ahead.

How a dashboard page reads

One filter bar

Periods (3M, 6M, 12M, This FY, All), dates, a financial year, and groups for Client and Work. A click on any chart is a removable chip.

The brief

What matters (flagged first), what changed (the biggest movers), and what to do next, one step per finding.

Two comparisons

Each figure shows its change against the same stretch of last year and against the whole of it. A switch chooses which leads.

Take a figure apart

Click a tile: by financial year, then month, then every dimension. The total always equals the tile.

The rows behind it

Every chart opens its rows in a drawer: sortable, searchable and exportable. Read-only.

Says how it is worked out

Each figure has a plain-language help: what it means and how it is calculated.

The Forecast tab: where a figure is heading

1 month | 3 months | 6 months | 12 months | 3 years

What it forecasts

- Orion: revenue, billed, cash collected, cost, margin
- Tally: billed, revenue, GST, documents, credit notes
- Narrow to a client, domain or case

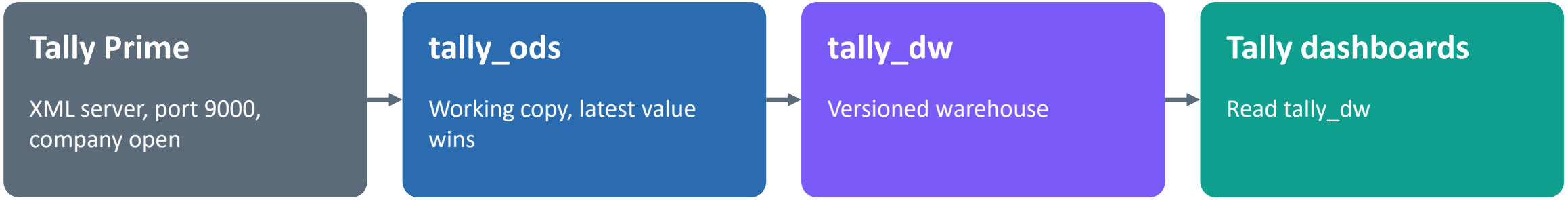
What it shows

- Expected total and a likely range
- Against the same length of time just gone
- A range that widens with distance
- Year totals: months done plus months to come

How it is honest

- Recent pace, growth on a year ago
- The usual pattern through the year
- How it was worked out and how well it would have done
- Pure Python, no extra dependency

Tally: the accountant's books, copied and reported



What the dashboards show

- Billing overview and activity
- Credit notes and income mix
- Clients and debtors
- A snapshot of the books (debtors, loans, cash)

The rules

- Read-only from Orion
- Loaded from the ETL page: working copy first, then the warehouse
- Fresh only as of the last Tally and warehouse loads
- Separate from Orion: never joined into one number

In the Copilot

- Eight read-only Tally tools
- Offered only to people with the Tally dashboard
- Answers say "in the Tally books"
- A question naming Tally is never answered from Orion

ETL: three pipelines, one page

Orion warehouse

orion_ods to orion_dw. Loads itself every hour, only if something changed. Run all or one table.

Tally working copy

Tally Prime on port 9000 to tally_ods. Started by a person from this page.

Tally warehouse

tally_ods to tally_dw. Refuses to start while the working copy is loading.

Run history

Started, finished, how long, person or schedule, and per-table counts and rows in the target.

One at a time

A second run is refused as "already running". A run cut off by a restart is closed on the next start.

Explain this failure

An AI button turns an error into plain words, the likely cause and what to try. It sees only the run name and errors.

Documents, letters and the Copilot

Keeping papers, writing on the letterhead, and asking in plain words.

The Document Center: a file explorer for company papers

Works like a computer

- A tree of company, clients, cases and folders
- Drag, move, rename, right-click, back and up
- Whole folders can be dropped in
- A folder deleted with what is in it says what would be lost

Versions and expiry

- Draft, Active, Superseded, Archived
- Any number of versions
- An expiry date with a reminder
- Expiring soon is a front-page filter

Finds words inside files

- Titles, tags and file names
- The text inside PDF, Word, Excel, PowerPoint
- Confidential documents need the right flag
- Everything is kept in the database, so it is in the backup

Filed by itself

- Each bill is filed when issued or changed
- Employee papers and EMD papers too
- Never fails the save that caused it
- Google Drive can be connected by an admin

Letterhead: anything, on the company paper

Letters, quotes, notices

Typed in a rich editor, saved, printed. "Write with AI" drafts from a few points or changes tone and language.

Stamp a PDF

Put a PDF made elsewhere (Word, Excel, Tally, a scan) onto the letterhead. Or print blank letterhead paper.

Bills and statements

A bill or a client's statement goes on the letterhead from Billing, drawn on the server.

The design

A header and a footer strip, replaced and positioned by an admin. Seven measurements place everything.

The AI Copilot: local first, tools only

1 Orion itself

Rules recognise common requests; tools work the answer out from real data, in plain words. No model.

2 A model on this computer

Ollama, for anything open-ended. A very small model only fills in a short form; Orion does the lookup.

3 A cloud model

Only if the server, the organisation's policy and the person all allow it. Names and long numbers are masked.

Only tools touch data

Each tool has a permission, a schema and, if it changes anything, a confirmation on screen. No tool runs SQL or acts on many records at once.

Confirm to change

It can create, change or delete records, export, make a PDF report, email a file, set a reminder and open a page. The person confirms.

Audited, never logged

Every action is audited. A question or an answer is never written to the route log. Keys never leave the server.

AI help inside five screens, and in chat

Screen	Button	What it does	What the model is given
Letter editor	Write with AI	Drafts from points, or rewrites; translates	The points or selected text
Outstanding, Send reminder	Reword with AI	Changes the tone of the reminder	Only the wording and "N bills, oldest D days": never a name or amount
Document upload	Suggest details	Proposes title, kind, dates, tags	The first 6,000 characters, or the file name
A record form	Fill from text	Proposes values from a pasted email	The pasted text and the allowed fields
ETL run	Explain this failure	Plain words: cause and what to try	The run name and its errors
Messages	Sparkle button	Rewrite, translate, suggest replies, summarise	This conversation only, as "Me" and "Them"

At most 20 AI jobs a person a minute. Private fields (Aadhaar, PAN, salary) are never offered to a model.

Tasks: a Jira-style tracker with personal to-do lists

Five ways to see work

- Board: a column per status, drag to move
- List with a filter on every column
- Sprint plan, calendar
- My work and personal to-do lists

A task has everything

- Status, assignee, reviewer, priority, type
- Dates, estimate, labels, history
- Comments, files, checklist, time, links
- Links to bills, cases and so on follow access

Rules that hold

- Company or Private projects
- Manager, Member, Viewer; admins manage all
- Finished only via a done status; reviews approve
- Two people never silently overwrite

Templates and repeats

- A template makes a task from a pattern
- A repeating task runs on schedule
- Only the latest missed run is made
- The bell and inbox tell people

Messages: chat, groups and calls

Conversations

- Presence, unread counts, search, archive, mute, pin
- Sent, read, failed with Retry
- Reply, react, forward, pin, edit, delete
- Files up to 12 MB, checked

Groups

- Up to 50 people
- Whoever makes it is its admin
- Names, members and roles managed in the group
- A group is a conversation like any other

Calls and voice notes

- Voice, video and screen sharing, one to one
- WebRTC, encrypted, browser to browser
- Only who, kind, outcome and length are kept
- Voice notes in any conversation

AI help

- Rewrite, translate, suggest replies, summarise
- Runs on this computer when it can
- Only this conversation is given to it
- Nothing said is ever logged

Appearance: each person chooses how Orion looks

Six themes

Default, Midnight, Aurora, Obsidian, Ivory and Flux, in light, dark or the device's setting.

Fine control

Accent colour, visual style, corner radius, density, interface size, higher contrast and reduced motion.

Classic or Lumen

Two interfaces over the same pages: the plain workspace, or glass, headlines and cards. Switch any time.

A welcome tour

Three steps for a new account: choose a look, see what your access opens, and four quick ways around.

Tools for administrators



SQL editor

A tree of the database, a query editor, results, saved queries and history. Admins only; read-only unless switched on; every run is audited.



History of a record

Who created, changed or deleted it, and each change field by field, newest first. Read from the warehouse.



Column filters

Every table column has a condition or a pick-of-values filter. Filters show as chips and combine.



Warehouse browser

Every column and every version of a record, with the surrogate keys, so a fact can be traced to its dimension version.



System page

Health, logs, database schema state, the last backup, and any error in the last hour. For admins.



Raise a request

Anyone can send a support request with attachments from the rail or the account menu.

Keeping it running, and getting it back



Backups

Daily, an encrypted dump of all four schemas, uploaded to a Backblaze B2 bucket. Uploaded documents and files are inside the database, so they are covered.



Health and logs

/api/health must answer ok; readiness says if the schema is wrong. Logs go to a file; a background job writes its result to its own table.



A connection pool

Opening a connection cost about 100 ms. With a pool a typical call went from 216 ms to 4 ms, and eight people at once no longer slow each other.



Safe schema and deploy

Every schema file is recorded, re-runnable and additive. A failed migration stops the deploy, and a failed smoke test rolls back.

How we know it works

1,656 backend tests

72 test files that need no database. DB-backed checks run only when their flag is set.

Contract tests

A form field, a warehouse column, a dashboard tab and a SQL file are each checked against the other side.

Shared test vectors

73 bill figures and 11 attendance figures that browser and server must both produce.

Check scripts

Themes readable, bill links, money fields, column filters, Lumen parity, bundle size. All must pass.

CI on every PR

Backend, frontend and a dependency audit. Packages are pinned and installed with constraints.

Change-impact map

A document lists what else breaks when something changes, so a small change does not surprise.

Where to read more

Document	What is in it
AGENTS.md	The short guide: layout, commands, and the rules that have caused real damage
docs/architecture.md	How the pieces fit: processes, schemas, data flow, authentication
docs/features.md	What each feature does, its files, tables and rules (the longest document)
docs/change-impact.md	What else breaks when you change something
docs/kt-money-to-dashboards.md	How money moves from a record to a dashboard figure
docs/money-flow.md	Controls, the bank, reconciliation, and an honest review of gaps
docs/backup.md, ai-setup.md	Backups and restore; setting up the Copilot
DEVELOPMENT.md, DEPLOYMENT.md	Branch, test, PR and release; dev versus production and rollback

What is not built yet, and what could come next

NOT IN THE SCOPE TODAY

- Double-entry accounting: Tally stays the book
- A month-end lock on past months
- The GST return itself (GSTR-3B)
- PF and ESI worked out from salary
- A bank feed: statements are uploaded
- A reconciliation of Orion against Tally
- A rehearsed restore drill

IDEAS FOR NEXT

- Automatic reminders: overdue bills and expiring documents, by email and WhatsApp
- Attendance on a phone, with offline entry
- A Hindi interface
- Month-end close, and an Orion-to-Tally reconciliation report
- More alerts: statement not uploaded, unmatched lines too old

Thank you

Questions, and a walk through the screens.

The written guides are in the docs folder.